Model training and conversion

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After completing the tutorial content of the dataset annotation, we can use the mainboard to start training the model.

```
This tutorial only introduces the model training and conversion with the CLI; you can refer to the official website yourself to modify the Python case
```

1. Model training

Use the CLI command to train the model directly

```
cd /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data
```

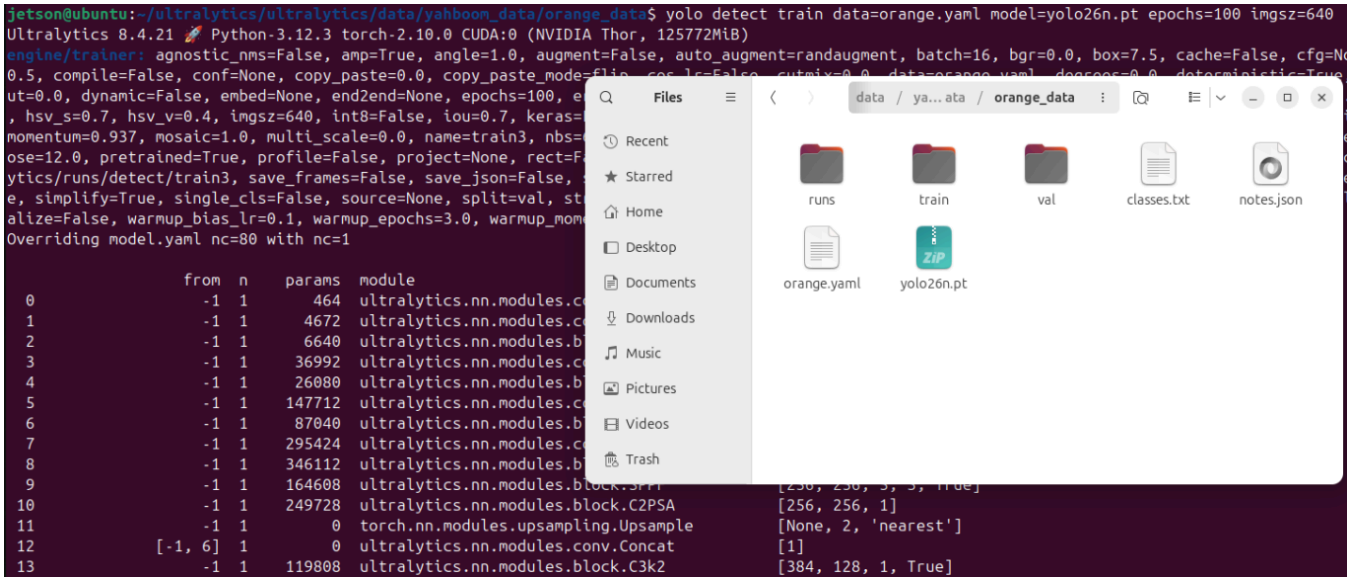
```
yolo detect train data=orange.yaml model=yolo26n.pt epochs=100 imgsz=640
```

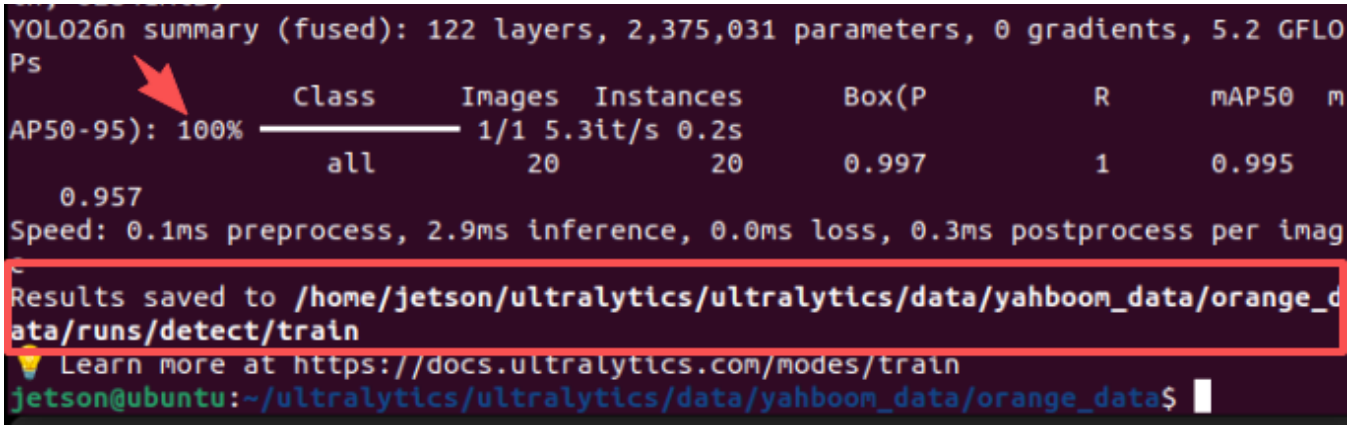
`data`: the dataset configuration file

`model`: the pre-trained model file

`epochs`: the number of training epochs

`imgsz`: the specified input image size





Finally, the printed model-related files after training are stored in the /home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train folder.

Note: each time the training is run, the number after train in the path automatically increases by 1; please record this path and use the new path of this model later.

2. Model conversion

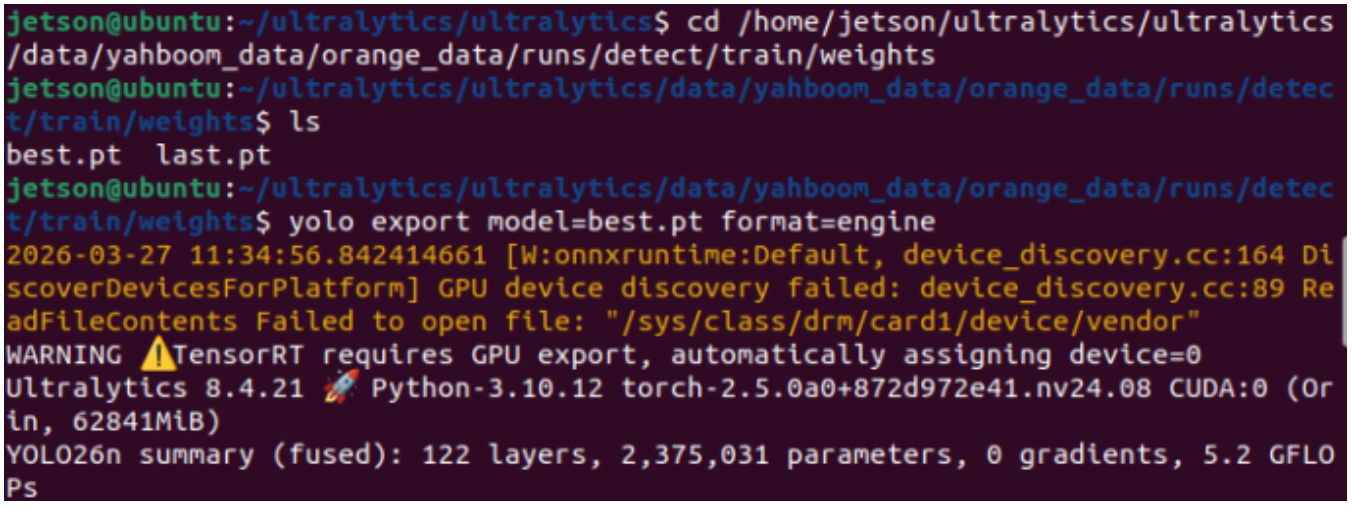
The final model is generated under the runs folder: generally the best.pt file is chosen for use

```
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs
```

Convert the model from the PyTorch format into TensorRT:

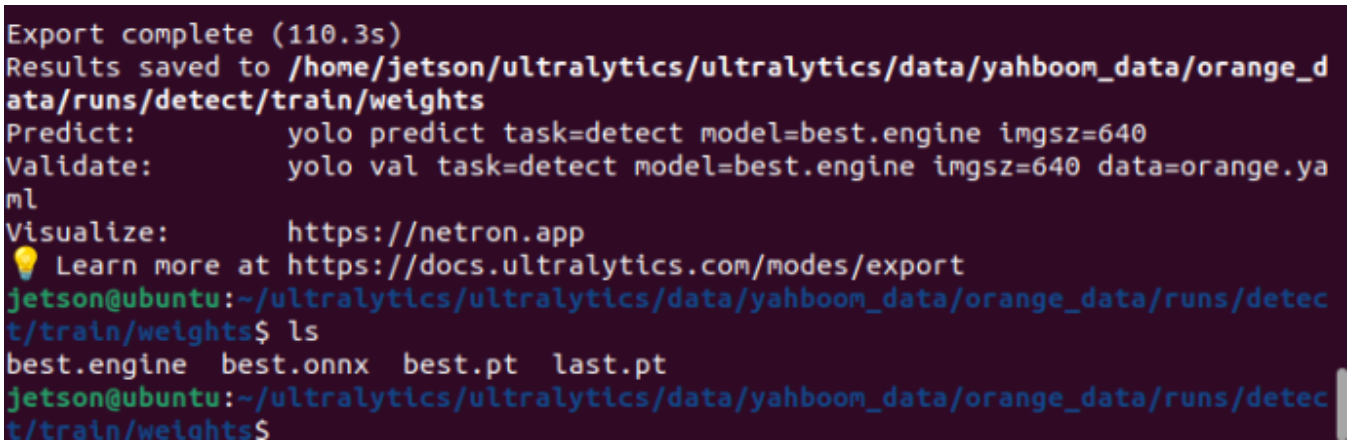
```
cd
/home/jetson/ultralytics/ultralytics/data/yahboom_data/orange_data/runs/detect/train/weights
ls
```

```
yolo export model=best.pt format=engine
```



Wait for the model conversion to complete; the best.engine and best.onnx files are generated in the current directory.

```
ls
```



References

https://docs.ultralytics.com/modes/train/